



PRIMARY 3 (GRADE 3) 2023 CONTEST PAPER

NAME: _____

Index Number:

SCHOOL: _____

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INSTRUCTIONS:

1. Please **DO NOT OPEN** the contest booklet until the Proctor has given permission to start.
2. TIME: **1 hour 30 minutes**.
3. There are 45 questions:
Section A: Questions 1 to 40 score 2 points each, no points are deducted for an unanswered question or wrong answer.
Section B: Questions 41 to 45 score 4 points each, no points are deducted for an unanswered or wrong answer.
4. Shade your answers neatly using a **2B lead pencil** in the Answer Entry Sheet.
5. PROCTORING: No one may help any student in any way during the contest.
6. No electronic devices capable of storing and displaying visual information are allowed during the course of the exam.
7. Strictly **No Calculators** are allowed into the exam.
8. All students must fill and shade their **Name, School and Index Number** in the Answer Entry Sheet and Contest booklet.
9. MINIMUM TIME: Students must stay in the exam hall for at least 1 hour.
10. A student must show detailed working and transfer answers to the Answer Entry Sheet.
11. **No exam papers and written notes can be taken out by any contestant.**

Appendix A

Instructions for completing your Answer Entry Sheet (AES)

1. Use only 2B lead pencils.
2. Write your name on both side of AS. On the front page of AES, please write your name and school name on the space provided.

NAME OF PARTICIPANT:

Henry Ong

SCHOOL NAME:

Ang Mo Kio Primary School

3. Write your index number in the space provided and fill in the proper circle directly below each number.

INDEX NUMBER:

0 6 5 2 2 0 0 2 5 3 7 9

0	1	2	3	4	5	6	7	8	9	0	1
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

4. Write your grade in the space provided and fill in the proper circle directly below your grade.

LEVEL (GRADE):

P1 (G1) P2 (G2) P3 (G3) P4 (G4) P5 (G5) P6 (G6)

☐ ☐ ☐ ☒ ☐ ☐

S1 (G7) S2 (G8) S3 (G9) S4 (G10) JC

☐ ☐ ☐ ☐ ☐

5. Mark only one answer to each question.
6. Make heavy black marks that fill the circle completely.

EXAMPLES OF SHADING**CORRECT****WRONG**

7. Write the answer in the space provided and fill in the proper circle directly below each number. There should only be one answer for each question.

Multiple Choice Answer

- 1 ☐ ☒ ☐ ☐ ☐
- 2 ☐ ☐ ☐ ☐ ☒
- 3 ☐ ☐ ☐ ☒ ☐
- 4 ☐ ☐ ☐ ☒ ☐
- 5 ☒ ☐ ☐ ☐ ☐
- 6 ☐ ☐ ☒ ☐ ☐
- 7 ☐ ☐ ☐ ☐ ☐
- 8 ☐ ☐ ☐ ☐ ☐
- 9 ☐ ☐ ☐ ☐ ☐
- 10 ☐ ☐ ☐ ☐ ☐
- 11 ☐ ☐ ☐ ☐ ☐
- 12 ☐ ☐ ☐ ☐ ☐
- 13 ☐ ☐ ☐ ☐ ☐
- 14 ☐ ☐ ☐ ☐ ☐
- 15 ☐ ☐ ☐ ☐ ☐

EXAMPLE for Questions 16 to 25
You must write and shade all 4 digits.

0 0 0 9

0	1	2	3	4
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

8. Erase cleanly any answer you wish to change.
9. Do not make stray marks on this AES.
10. Do not fold or staple the AES.

Section A: 40 Open-ended Questions (40 × 2 marks = 80 marks)

Read the questions carefully. For questions 1 to 40, show your working clearly and write your answers in the blank provided. Then write and shade your answers in the Answer Entry Sheet (AES) sheet provided.

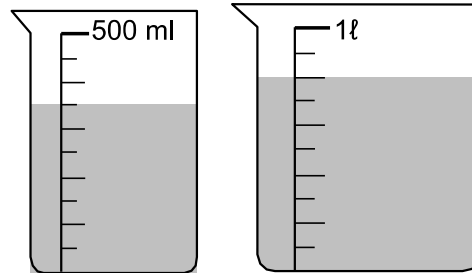
1. 2 tens more than is 5860.
What is the missing number in the box?

Answer: _____

2. $18 + 6 + 6 = \boxed{?} \times 6$
What is the missing number in the box?

Answer: _____

3. What is the total volume of water in the 2 beakers below?
Give your answer in ml.

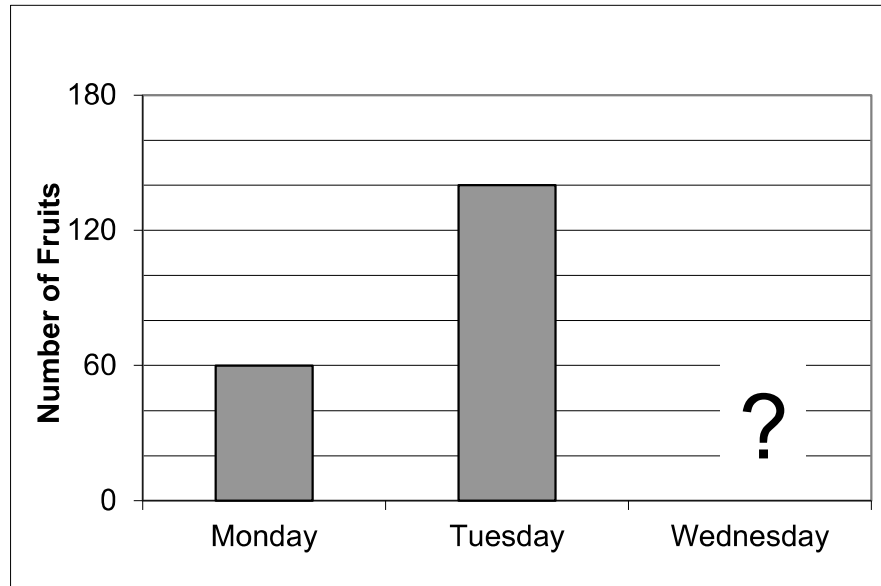


Answer: _____ ml

4. Express 3 h 25 min in minutes.

Answer: _____ min

5. The graph below shows the number of fruits James sold.



How many fruits must he sell on Wednesday so that he can sell a total of 280 fruits for the 3 days?

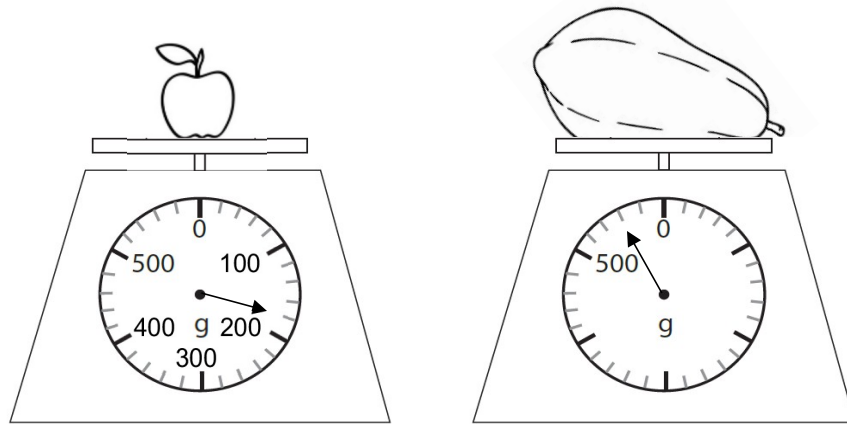
Answer: _____

6. What is the missing number in the box below?

$$\frac{21}{49} = \frac{3}{\boxed{}}$$

Answer: _____

7. Study the diagram below.
How much heavier is the papaya than the apple?
Give your answer in grams.



Answer: _____ g

8. Which one of the following fractions is greater than $\frac{1}{2}$?

Option (1) $\frac{3}{6}$

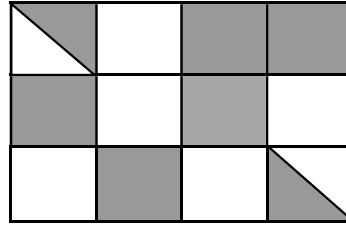
Option (2) $\frac{3}{7}$

Option (3) $\frac{4}{7}$

Option (4) $\frac{4}{9}$

Answer: Option (_____)

9. What fraction of the figure below is shaded?

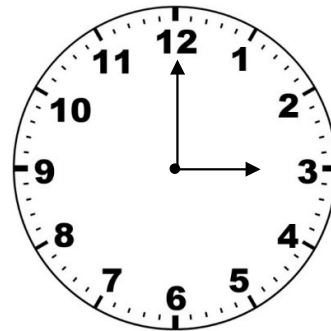


- Option (1) $\frac{5}{12}$
Option (2) $\frac{6}{12}$
Option (3) $\frac{7}{12}$
Option (4) $\frac{8}{12}$

Answer: Option (_____)

10. The clock below is 20 min fast.
Which of the following options shows the actual time?

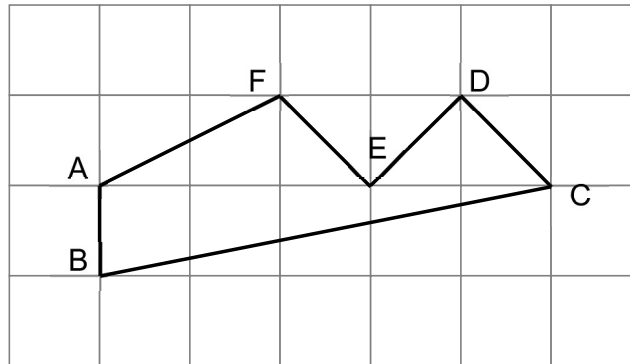
- Option (1) 20 minutes to 3
Option (2) 20 minutes past 3
Option (3) 20 minutes to 2
Option (4) 20 minutes past 2



Answer: Option (_____)

11. Look at the diagram below.

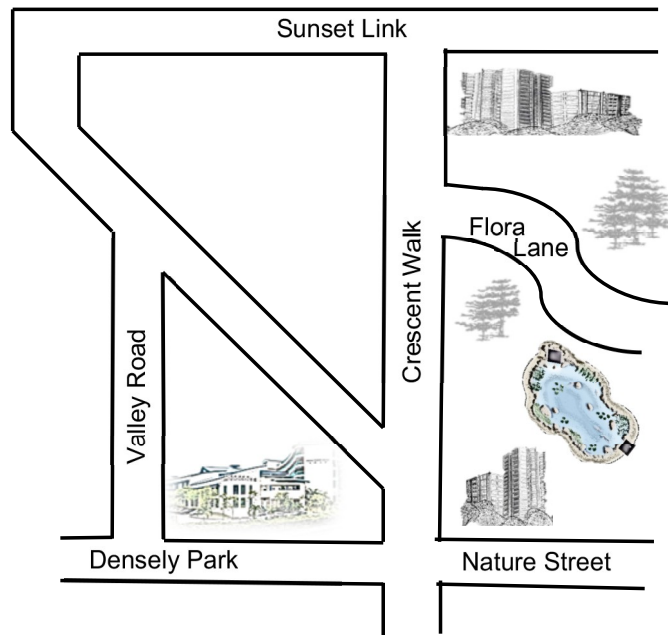
Which of the following options shows a pair of perpendicular lines?



- Option (1) AB and CD
- Option (2) CD and DE
- Option (3) AF and EF
- Option (4) EF and CD

Answer: Option (_____)

12. Study the neighbourhood map below.



Which of the following options is a correct statement?

- Option (1) **Nature Street** is parallel to **Flora Lane**.
- Option (2) **Crescent Walk** is parallel to **Valley Road**.
- Option (3) **Nature Street** is parallel to **Densely Park**.
- Option (4) **Sunset Link** is parallel to **Crescent Walk**

Answer: Option (_____)

13. Tina bought a bag.

She gave the cashier 2 fifty-dollar notes.

She received \$2.75 change.

Which of the following options shows the cost of the bag?

Option (1) \$98.75

Option (2) \$98.25

Option (3) \$97.75

Option (4) \$97.25

Answer: Option (_____)

14. Julie saves only five-cent and twenty-cent coins in her piggy bank.

She wants to buy a birthday present that costs \$6.

She has only \$4.85 in the piggy bank.

Which of the following options shows the additional coins she will need in order to buy the birthday present?

Option (1) 1 five-cent coins and 5 twenty-cent coins

Option (2) 5 five-cent coins and 3 twenty-cent coins

Option (3) 7 five-cent coins and 4 twenty-cent coins

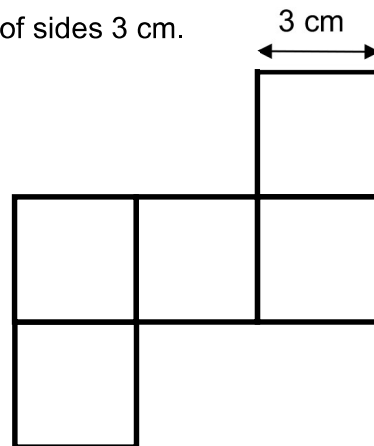
Option (4) 8 five-cent coins and 5 twenty-cent coins

Answer: Option (_____)

15. Mr Mok is a taxi driver.
He drove 6 h on Sunday.
He drove 2 h 55 min less on Monday than on Sunday.
What is the total amount of time he drove on Sunday and Monday?
Give your answer in minutes.

Answer: _____ min

16. The figure below is made up of 5 identical squares of sides 3 cm.
What is the area of the figure?
Give your answer in square centimetres.

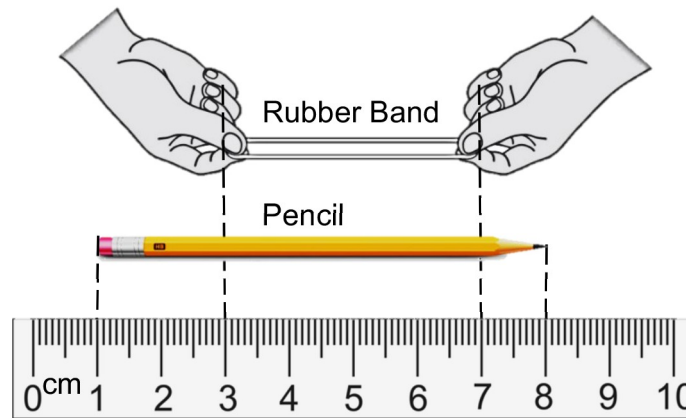


Answer: _____ cm^2

17. Study the picture below.

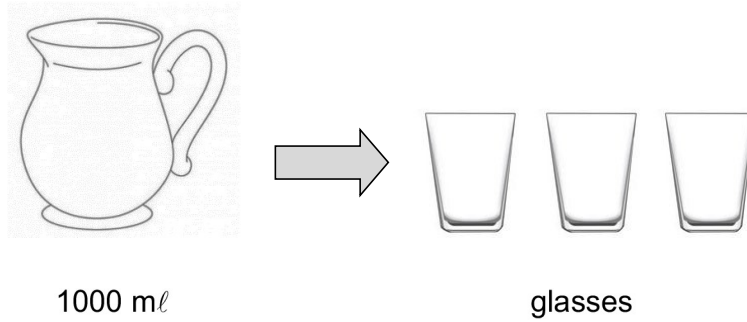
How long more must the rubber band be stretched so that it has the same length as the pencil?

Give your answer in centimetres.



Answer: _____ cm

18. A jug holds 1000 ml of orange juice.
After filling 3 identical glasses, 376 ml of orange juice is left in the jug.
What is the capacity of one such glass?
Give your answer in millimetres.



Answer: _____ ml

19. A farmer had 450 apples.
He put them into 7 boxes equally and had 58 apples left.
How many apples were there in each box?

Answer: _____

20. Sticks were used to form triangles as shown in the figures below.



Figure 1
3 sticks

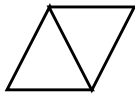


Figure 2
5 sticks

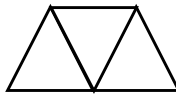


Figure 3
7 sticks

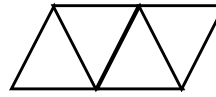


Figure 4
9 sticks



Figure 9

The arrangement continued in the same pattern.

How many sticks would be needed to form Figure 9?

Answer: _____ sticks

21. Mangoes were sold at \$24 for 3 kg.

Mdm Ong sold 27 kg of mangoes.

How much did she receive from selling the mangoes?

Answer: \$ _____

22. John, Sam, and Tom are standing in a row to take a photograph.
They will switch positions after each shot.
How many different ways can they stand in a row to take the photograph?



Answer: _____

23. The total mass of 3 mangoes and 1 watermelon is 2 kg 600 g.
The mass of each mango is 350 g.
What is the mass of the watermelon?
Give your answer in grams.

Answer: _____ g

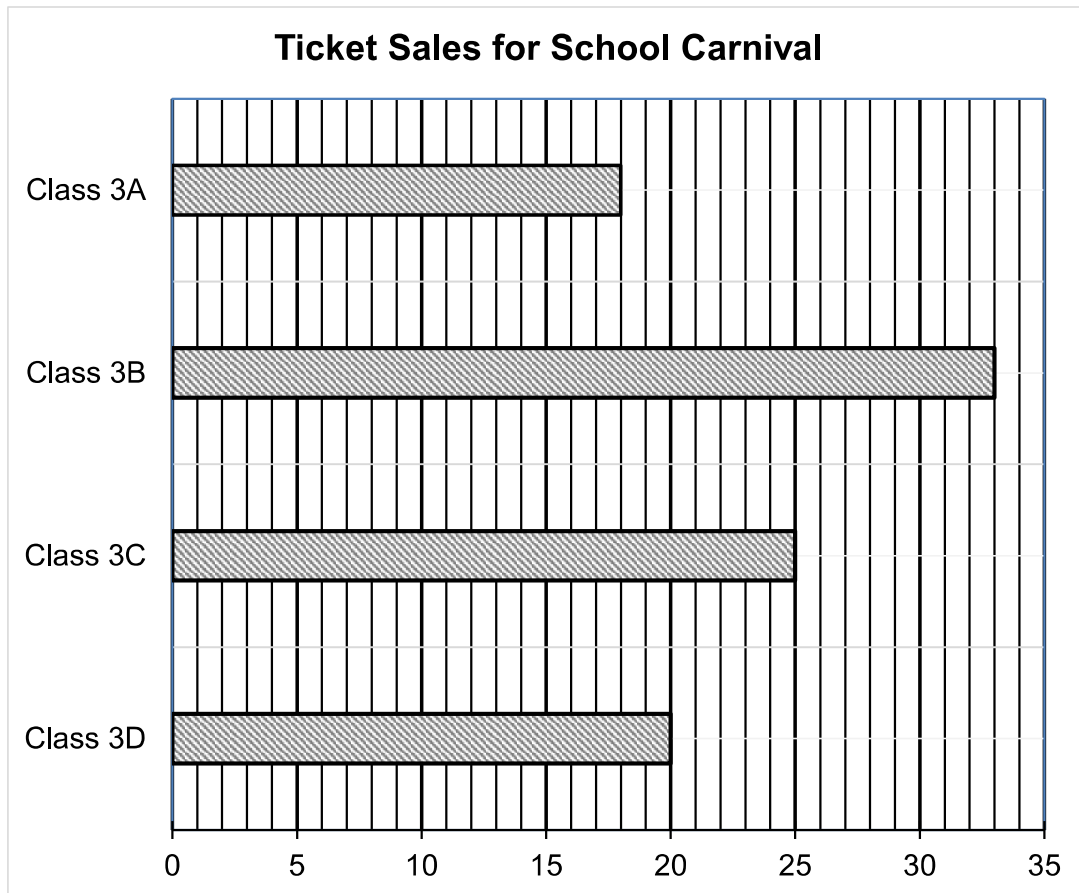
24. A wallet and 4 pouches cost \$265.
The wallet and 2 such pouches cost \$189.
Find the cost of 1 wallet

Answer: \$ _____

25. Polly has 72 oranges.
She has 8 times as many oranges as apples .
She has 4 times as many pears as apples.
How many pears does Polly have?

Answer: _____

26. The bar graph below shows the number of carnival tickets sold by four classes in Primary Three.



Each carnival ticket cost \$8.

What is the difference between the class with the most amount of money collected and the class with the least amount of money collected?

Answer: \$ _____

27. Jack and Tom had a total of \$330.

Jack gave \$87 to Tom.

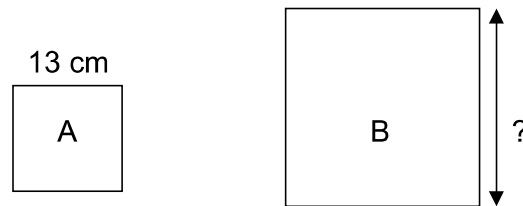
In the end, Tom had twice as much money as Jack.

How much money did Tom have at first?

Answer: \$ _____

28. A wire is 2 m long.

It is cut into 2 pieces which are bent to form two squares, A and B, as shown below.

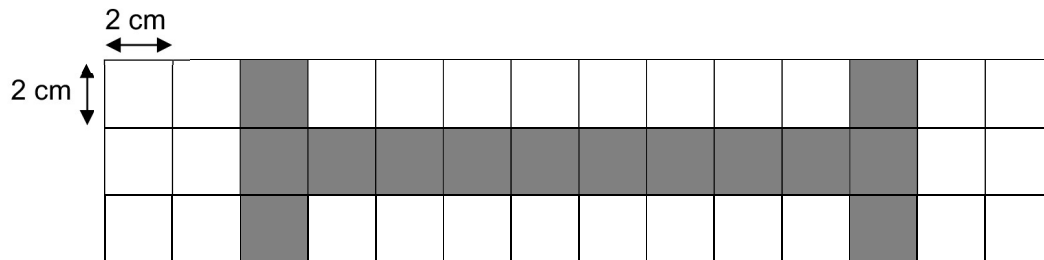


What is the length of each side of Square B?

Give your answer in centimetres.

Answer: _____ cm

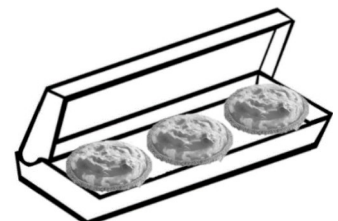
29. Find the perimeter of the shaded figure.
Give your answer in centimetres.



Answer: _____ cm

30. At a bakery, cheese tarts are sold in boxes of 3.
Each box of cheese tart costs \$8.
Each piece of cheese tart costs \$2.80
Mdm Sim has \$55.
What is the greatest number of cheese tarts she can buy?

Cheese Tarts



3 for \$8
1 for \$2.80

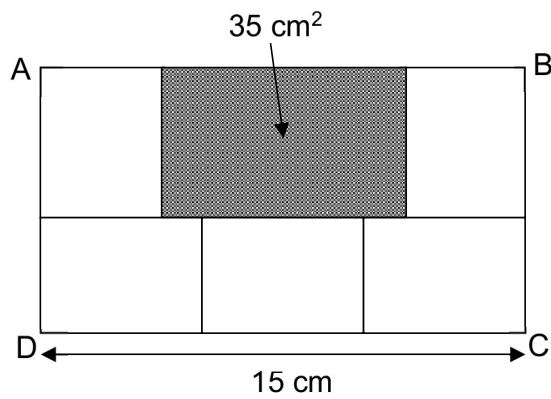
Answer: _____

31. Rectangle ABCD is made up of 1 big shaded rectangle and 5 identical smaller unshaded rectangles.

The area of the shaded rectangle is 35 cm^2 and the length of DC is 15 cm.

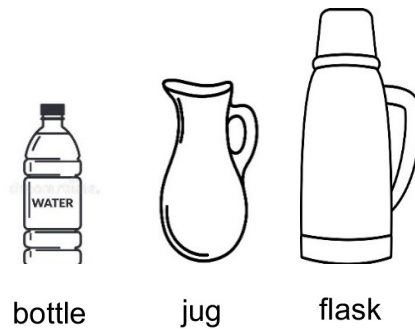
What is the area of Rectangle ABCD?

Give your answer in square centimetres.



Answer: _____ cm^2

32. A jug can hold 1050 ml of water.
A bottle can hold 225 ml less water than the jug.
A flask can hold 3 times as much water as the bottle.
How much water can the items hold altogether?
Give your answer in millilitres.



Answer: _____ ml

33. A container with 20 identical beads weighed 826 g.
When 7 beads were removed, the mass of the container with the remaining beads was 721 g.
What was the mass of 9 such beads?
Give your answer in grams.

Answer: _____ g

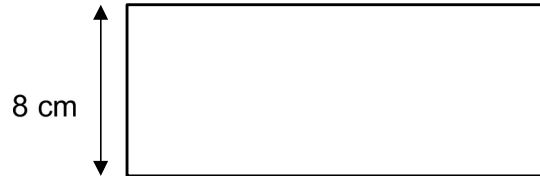
34. Look at the diagram below.

The length of a rectangle is 3 times its breadth.

Its breadth is 8 cm.

What is the perimeter of the rectangle?

Give your answer in centimetres.



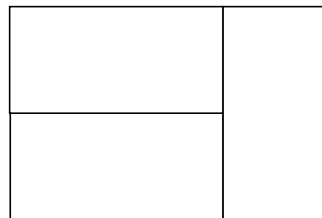
Answer: _____ cm

35. The figure below is formed using 3 identical rectangles.

The perimeter of the figure is 60 cm.

Find the total area of the figure.

Give your answer in square centimetres.

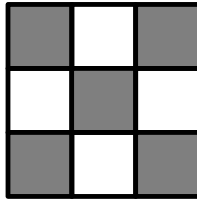


Answer: _____ cm²

36. Jay and Kenny had an equal amount of money.
Jay bought 5 cookies and had \$3.60 left.
Kenny wanted to buy 8 cookies but he was short of \$4.80.
Find the cost of 1 cookie.
Give your answer in cents.

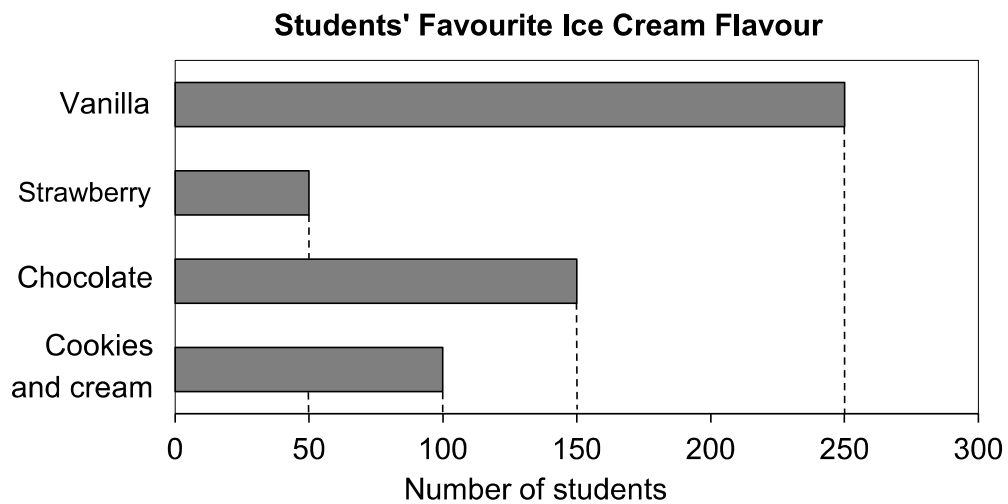
Answer: _____ cents

37. The figure below is made up of 9 identical squares.
5 of the squares are shaded.
How many unshaded squares should be shaded so that $\frac{2}{3}$ of the figure is shaded?



Answer: _____

38. A total of 480 students was asked to choose their favourite ice cream flavours. The bar graph shows the numbers for each of the 4 Ice Cream flavours.



How many students have chosen more than one flavours?

Answer: _____

39. Rita and Sarah had the same number of beads.
Rita gave away 35 beads and Sarah received 40 beads.
In the end, Sarah had 4 times as many beads as Rita.
How many beads did Sarah have at first?

Answer: _____

40. A farmer had 98 oranges and pears.
He packed all of them into similar boxes.
Each box can hold 6 oranges or 5 pears.
There are 2 more boxes of pears than oranges.
How many oranges did he have?

Answer: _____

Section B: 5 Open-ended Questions (5 × 4 marks = 20 marks)

Read the questions carefully. For questions 41 to 45, show your working clearly and write your answers in the blank provided. Then write and shade your answers in the Answer Entry Sheet (AES) sheet provided.

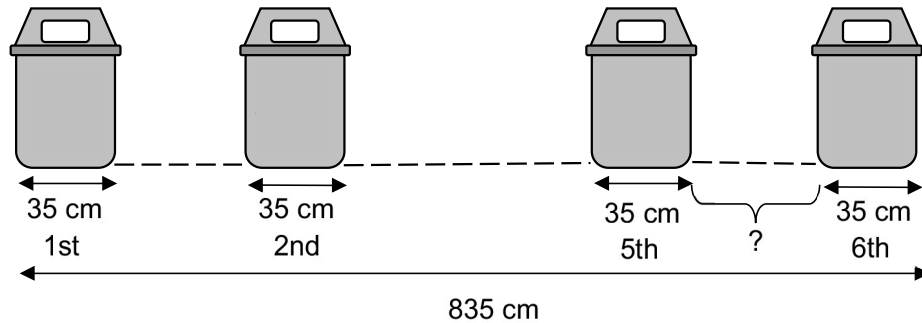
41. A shopkeeper has 622 pens of 3 different colours in a shop.
There are 4 times as many blue pens as green pens.
There are 35 fewer red pens than blue pens.
How many blue and red pens are there altogether in the shop?

Answer: _____

42. 11 taxis were hired to fetch 52 passengers to their hotels.
Each small taxi carried 4 passengers and each big taxi carried 6 passengers.
How many small taxis were hired?

Answer: _____

43. 6 dustbins are lined up in a row with equal distance between them.
The length of each dustbin is 35 cm.
The total distance from the first dustbin to the last dustbin is 835 cm.
What is the distance between the 5th and 6th dustbins?
Give your answer in centimetres.



Answer: _____ cm

44. Mrs Tan invited some children to her party.
The number of girls was twice the number of boys at the party.
Each girl received 2 lollipops and each boy received 3 lollipops.
Mrs Tan gave out 154 lollipops in all.
How many children were invited to the party?

Answer: _____

45. Some stars are arranged in rows.
The first 3 rows are shown below.



Which row has a total of 947 stars?

Answer: Row _____

~~ End of Grade 3 Paper ~~